

Selective Attention Merging for low resource tasks: A case study of Child ASR

Natarajan Balaji Shankar, Zilai Wang, Eray Eren, Abeer Alwan

Department of Electrical and Computer Engineering

University of California Los Angeles

Los Angeles, USA

{balaji1312, zilaiwang2001, erayeren}@ucla.edu, alwan@ee.ucla.edu

Abstract—While Speech Foundation Models (SFMs) excel in various speech tasks, their performance for low-resource tasks such as child Automatic Speech Recognition (ASR) is hampered by limited pretraining data. To address this, we explore different model merging techniques to leverage knowledge from models trained on larger, more diverse speech corpora. This paper also introduces Selective Attention (SA) Merge, a novel method that selectively merges task vectors from attention matrices to enhance SFM performance on low-resource tasks. Experiments on the MyST database show significant reductions in relative word error rate of up to 14%, outperforming existing model merging and data augmentation techniques. By combining data augmentation techniques with SA Merge, we achieve a new state-of-the-art WER of 8.69 on the MyST database for the Whisper-small model, highlighting the potential of SA Merge for improving low-resource ASR.

Index Terms—Automatic Speech Recognition, Speech Foundation Models, Model Merging, Children’s Speech

I. INTRODUCTION

Recently, Speech Foundation Models (SFMs) have increasingly come to dominate the landscape of Automatic Speech Recognition (ASR) [1]–[9] due to their impressive performance in a range of different speech datasets, and considerable zero-shot ability. The success of SFMs can be attributed to several factors, including large-scale pretraining on diverse datasets and learning objectives that leverage unlabeled (self supervised) or weakly supervised data. However, the performance of these models in child speech related tasks still lags behind that seen in general adult speech [10]. This performance gap can be primarily attributed to the significant acoustic and linguistic differences between child and adult speech, such as higher pitch, greater variability in pronunciation, and the use of simpler vocabulary and sentence structures [11].

Several strategies have been proposed to tackle this domain mismatch. Signal processing based acoustic techniques to address the data scarcity issue involve using data augmentation techniques [12]–[15] to artificially increase the variety of data seen during fine-tuning of the model. These techniques involve applying transformations to the original speech signals, such as pitch shifting, time stretching, or masking parts of the spectrogram, to create new training examples that can help the model generalize better to unseen child speech. Another way to increase the diversity of training data seen is to use synthetic data created using voice conversion or TTS on in-domain text to increase the robustness of models to child speech [16]–[18]. While these methods do improve the performance of SFMs in the new domain, these often involve creating artificial data, before retraining the model on each new dataset encountered.

Model merging [19]–[24] has recently emerged as a compelling alternative to training models with new augmented sets. By leveraging

the knowledge learned by a model on more comprehensive corpora, we can improve the performance of the same model on a more limited dataset. This approach avoids the need for extensive data collection and retraining, which can be particularly beneficial in scenarios like low-resource child ASR. In addition, model merging techniques have been shown to increase performance on in-domain tasks by increasing the generalizability of the model [19]. This suggests that model merging can both help models adapt to new domains and improve performance on existing domains without requiring explicit fine-tuning, making it a promising approach for addressing the domain mismatch in child ASR.

This paper presents an investigation into the application of model merging in low-resource child ASR, along with introducing a novel Selective Attention (SA) Merge technique. Our contributions can be summarized as follows:

- Exploring the viability of existing model merging techniques for low-resource child ASR across different SFMs
- Introduction of a new SA Merge technique for low-resource domain adaptation of SFMs
- Adaptation of existing data augmentation techniques to different child speech datasets through task vector transfer

The remainder of this paper is organized as follows. Section II introduces the proposed SA Merge method, and serves as an introduction to other model merging methods tested. Experimental setups and datasets are described in Section III. Results are discussed in Section IV, and we conclude the paper in Section V.¹

II. METHODS

A. Model Merging

Model merging has emerged as a promising research area, aiming to combine multiple domain specific models into a single model. However, its effectiveness for child ASR remains unexplored. We present the first evaluation of these methods for merging models fine-tuned for child ASR, investigating the following techniques:

- **Linear Interpolation (Lerp)** [19]: Lerp creates a merged model by computing a weighted average of the parameters from individual models.
- **Spherical Linear Interpolation (Slerp)**: Similar to Lerp, Slerp performs interpolation in a spherical space [25], often resulting in smoother transitions between model parameters.
- **Task Arithmetic (TA)** [22]: TA involves computing task-specific vectors by taking the difference between task-specific models and a pretrained base model. The vectors are combined

¹Our code, models, and data splits are available at https://github.com/balaji1312/sa_merging

using predefined scaling factors to adjust the relative importance of different models during the merging process.

- **Regression Mean (RegMean)** [23]: RegMean merging formulates model merging as an optimization problem, minimizing prediction differences between the merged model and the individual models through linear regression.
- **TIES Merging (TIES)** [24]: TIES Merging tackles conflicts in model merging by trimming low-magnitude parameters, resolving sign disagreements, and merging parameters with consistent signs.
- **DARE Merging (DARE)** [21]: DARE can be used in combination with other merging methods, as it involves dropping a random percent of parameter differences, and rescaling the remaining weights before merging.

B. Selective Attention Merge

Previous analyses have shown the importance of attention maps in different layers for speech-based learning, especially for speech in low-resource [26] and noisy [27] environments. Building on these insights, we propose a novel approach called Selective Attention (SA) Merge, which focuses on merging the task vectors of attention matrices while preserving the weights of other layers.

Drawing inspiration from [28] on cross-modal merging techniques and [29] on model merging within Hidden Markov Models (HMMs) for low-resource domains, we propose a novel approach to transfer knowledge from a comprehensive source domain to the low-resource target domain. While [29] focused on merging speech models using HMMs, our approach centers around the merging of attention matrices. SA Merge combines the task vectors of attention matrices from two models, $\mathcal{M}_1$ (fine-tuned on child speech) and $\mathcal{M}_2$ (fine-tuned on diverse adult speech), as follows:

$$\mathcal{M}_{SA} \tau_i^{Q,K,V} = \lambda_i \cdot \mathcal{M}_1 \tau_i^{Q,K,V} + (1 - \lambda_i) \cdot \mathcal{M}_2 \tau_i^{Q,K,V} \quad (1)$$

where $\mathcal{M}_{SA} \tau_i^{Q,K,V}$ represents the merged task vectors for the query, key, and value matrices in the i -th attention layer of the new model $\mathcal{M}_{SA}$. $\mathcal{M}_1 \tau_i^{Q,K,V}$ and $\mathcal{M}_2 \tau_i^{Q,K,V}$ are the corresponding task vectors from the child speech and adult speech models respectively, and λ_i is the mixing ratio for the i -th layer, controlling the contribution of each model.

The mixing ratio λ_i is further defined as:

$$\lambda_i = \lambda^{\alpha_i} \quad (2)$$

where λ is a base mixing factor and α_i is an exponent that controls the rate of change of the mixing ratio across layers.

By weighting the mixing ratio in an exponential manner, we aim to give higher importance to the lower layers from the child speech model $\mathcal{M}_1$. This is motivated by the assumption that lower layers capture more acoustic and phonetic features, which are crucial for distinguishing child speech from adult speech. At higher layers, the influence of the adult speech model $\mathcal{M}_2$ gradually increases, allowing the merged model to benefit from the broader linguistic knowledge captured in the adult speech data. Unlike [28], where model merging is explored between attention matrices from models trained on different modalities, we apply task vector based merging, taking into account the amount of target domain data learned by each model through an exponential weighting. For non-attention layers, we retain the weights from the target domain fine-tuned model, i.e., the child speech model $\mathcal{M}_1$. This ensures that the merged model retains the specialized knowledge acquired from the child speech data, while benefiting from the capabilities of the adult speech model $\mathcal{M}_2$.

C. Task Vector Transfer

Recent advancements in applying task vectors to speech have demonstrated remarkable success in transferring learned augmentations across datasets [30]. This process typically involves calculating the difference between task vectors derived from two models: $\mathcal{M}$ trained on the target dataset $\mathcal{D}$, and $\mathcal{M}'$ trained on an augmented version of the dataset $\mathcal{D}'$.

$$\tau_i = \theta_{i,1} - \theta_{i,2} \quad \forall \theta_{i,1} \in \mathcal{M}, \quad \theta_{i,2} \in \mathcal{M}' \quad (3)$$

where τ_i is the calculated task vector i -th attention layer of the new model, and $\theta_{i,1}$ and $\theta_{i,2}$ are the corresponding parameters from $\mathcal{M}$ trained on $\mathcal{D}$, and $\mathcal{M}'$ trained on $\mathcal{D}'$.

We extend this concept by investigating the transferability of these learned task vectors to models fine-tuned on different child speech datasets. Our goal is to assess whether the performance gains observed on the source dataset can be replicated in a new setting without any additional data augmentation. Furthermore, recognizing that the vector difference operation may not entirely eliminate all characteristics learned from the source dataset $\mathcal{D}$, we also evaluate the performance of these task vectors in a zero-shot setting to provide insights into the extent to which the learned task vectors capture generalizable knowledge about child speech patterns.

III. EXPERIMENTS

A. Experimental Setup

To assess the effectiveness of various merging techniques, including our proposed SA Merge, we conducted experiments across a diverse range of supervised and self-supervised speech foundation models (SFM). Specifically, we evaluated several models from the Whisper family [6] of varying sizes, using the Hugging Face Transformers [31] library for fine-tuning. Among SSL models, we evaluated the base versions of Wav2Vec2.0 [3], HuBERT [2] and WavLM [1]. All SSL models were trained with an identical character level CTC loss based on their implementation in the fairseq toolkit [32]. To offer a comparison of model merging techniques, we evaluate Lerp, Slerp, TA, RegMean, TIES, and DARE, in addition to our proposed SA Merge. The hyperparameters α and λ for SA Merge were tuned within the ranges [0.7, 0.9] and [0.1, 0.3] respectively. Our implementation of SA Merge, as well as the evaluation of the different models, was facilitated by the Mergekit library [33]. All models listed were fine-tuned using 2 Nvidia A6000 GPUs.

B. Datasets

For our merging experiments, we fine-tune two models: $\mathcal{M}_1$ on low-resource child speech and $\mathcal{M}_2$ on mainstream speech. To obtain $\mathcal{M}_2$, we train a given pretrained base model on the train-100-hour subset of the LibriSpeech (LS) corpus [34]. For $\mathcal{M}_1$, we utilize the following datasets:

1) *My Science Tutor*: The My Science Tutor (MyST) corpus [35] comprises approximately 240 hours of transcribed conversational children's speech, spanning grades 3 to 5, collected during virtual tutoring sessions on subjects including physics, geography, biology, and other science topics. Similar to [36], we identify and filter low quality audio samples by removing utterances with WER larger than 50% (after passing through Whisper-large-v2) or with less than 3 words are removed. Utterances longer than 30s are also removed in both the training and test sets, resulting in filtered data splits as follows: train (133h), dev (21h), and test (25h). To verify the efficacy of the proposed method under more constrained settings, we also separately prepare 1-hour, 5-hour, and 10-hour subsets of the MyST train corpus.

2) *CMU Kids*: To demonstrate the transferability of task vectors across different children’s speech datasets, we evaluated performance on the CMU Kids Corpus [37]. This corpus consists of 5180 utterances of read speech from 76 speakers, totaling 9 hours of child speech data. The utterances are randomly partitioned into train (70%), development (15%), and test (15%) sets, ensuring no speaker overlap between the sets.

3) *Data Augmentations*: To provide a contrast to the use of model merging, we compare several widely employed data augmentation methods on the MyST corpus. These include pitch perturbation (PP) [12], speed perturbation (SP) [13], vocal tract length perturbation (VTLP) [14], and SpecAugment (SpecAug) [15]. In addition to the above methods, we also generate synthetic TTS data on the MyST corpus.

- **Pitch perturbation (PP)** The fundamental frequency of each utterance is randomly shifted up or down by 1 to 12 semitones, creating two additional copies.
- **Speed perturbation (SP)** The speed of each utterance is modified, creating two copies with perturbation rates of 0.9 and 1.1.
- **Vocal tract length perturbation (VTLP)** This technique applies frequency warping to the speech signal, creating two copies with perturbation rates of 0.9 and 1.1.
- **SpecAugment (SpecAug)** Random masking of consecutive frequency bands and time frames is applied while training.
- **Synthetic Data**. Synthetic data is generated using StyleTTS 2 [38] with cross-utterance text, doubling the training data. To avoid contamination, only speakers in the respective training subset are used for synthetic data generation.

IV. RESULTS

A. Can Model Merging facilitate Knowledge Transfer in Low Resource Child ASR?

1) *Evaluation of Model Merging techniques on Supervised SFMs*: We first examine the effectiveness of different merging techniques to facilitate knowledge transfer using the Whisper-small model. For this purpose, we combine a model fine-tuned on the 100-hour subset of the LibriSpeech (LS) corpus [34] with models fine-tuned on the 1-hour, 5-hour, 10-hour, and the full train subset of the MyST corpus [35]. We conduct a hyperparameter search for each model merging method and report the best results in Table I. Our results demonstrate that the proposed SA Merge technique provides the most reliable improvements in reducing Word Error Rate (WER) across various data subsets. All subsequent results in this section show a statistically significant ($p < 0.05$) improvement for SA Merge compared to the baseline zero-shot performance of the models.

Next, we examine the impact of model merging when the size of the supervised SFM is varied. We train supervised SFMs of different sizes from the Whisper family on the 1-hour, 5-hour, 10-hour, and the full train subset of the MyST corpus. We then analyze the effect of merging these models with a model fine-tuned on the LS train-100 subset in Table II. These tests are performed using the best-performing method from Table I (SA Merge), but we note that similar trends are observed with other common merging methods. Generally, we find that as the available training data decreases or the model size shrinks, model merging enhances model robustness. Notably, SA Merge achieves a 14% reduction in relative WER on a Whisper-base model fine-tuned on the MyST train 1-hour subset, highlighting its efficacy in extremely low-resource scenarios.

TABLE I

WER RESULTS ON MYST TEST SET FROM MERGING WHISPER-SMALL MODELS TRAINED ON MYST AND LS TRAIN-100. ZERO SHOT DENOTES A MODEL WITHOUT FINE-TUNING. FT DENOTES A MODEL FINE-TUNED ON THE RESPECTIVE MYST SUBSET WITHOUT ANY MERGING. BOLD FACE NUMBERS INDICATE BEST RESULTS.

Merging Method	MyST test WER			
	MyST 1hr	MyST 5hr	MyST 10hr	MyST full
Zero Shot	13.44			
ft	10.64	10.05	9.94	9.34
Lerp	10.51	9.94	10.05	8.86
Slerp	10.51	9.94	10.05	8.88
TA	10.60	10.03	10.11	9.10
DARE + TA	10.43	10.07	10.16	9.16
TIES	10.70	9.97	10.00	8.92
SA Merge (Ours)	10.40	9.85	9.80	8.85

TABLE II

WER RESULTS ON MYST TEST SET FROM MERGING SUPERVISED SFMS TRAINED ON MYST SUBSETS AND LS TRAIN-100 USING SA MERGE. BOLD FACE NUMBERS INDICATE BEST RESULTS.

Model	Merging Method	MyST test WER			
		MyST 1hr	MyST 5hr	MyST 10hr	MyST full
Whisper tiny	ft	16.12	15.23	14.36	11.63
	SA Merge	15.26	14.38	14.00	11.52
Whisper base	ft	14.95	12.62	12.15	10.33
	SA Merge	12.84	12.30	11.55	9.87
Whisper medium	ft	9.92	9.42	9.19	8.86
	SA Merge	9.82	9.28	9.10	8.63
Whisper large v3	ft	11.31	9.54	9.42	9.12
	SA Merge	9.41	9.14	9.03	8.74

2) *Comparison with other Data Augmentation Techniques*: Table III presents a comparison of common data augmentation techniques applied to various MyST subsets. As data augmentation necessitates retraining the model with the newly augmented data, whereas SA Merge operates on existing models, we also investigate the potential benefits of combining these two approaches. Our results indicate that across all data augmentation methods, the application of SA Merge consistently improves model performance on the MyST test set. Notably, utilizing SA Merge in conjunction with SpecAug yields a WER of 8.69, establishing a new state-of-the-art performance on the MyST dataset for the Whisper-small model.

TABLE III

WER RESULTS ON MYST TEST SET FROM MERGING WHISPER-SMALL MODELS TRAINED ON AUGMENTED MYST DATA AND LS TRAIN-100 USING SA MERGE. PP, SP, VTLP, SPEC AUG DENOTE DATA AUGMENTATIONS. TTS AND REAL INDICATE PURE SYNTHETIC TTS DATA AND ORIGINAL MYST DATA. BOLD FACE NUMBERS INDICATE BEST RESULTS.

Augmentation/ Merging Method	MyST test WER			
	MyST 1hr	MyST 5hr	MyST 10hr	MyST full
PP	10.21	9.99	9.62	8.84
PP + SA Merge	9.65	9.29	9.32	8.80
SP	10.04	10.72	10.38	8.89
SP + SA Merge	9.53	9.52	9.14	9.01
VTLP	9.91	9.64	9.18	8.95
VTLP + SA Merge	9.63	9.24	9.04	8.75
SpecAug	9.77	9.35	9.25	9.03
SpecAug + SA Merge	9.54	9.21	9.05	8.69
TTS	13.23	12.04	12.19	12.61
TTS + Real	9.13	9.25	9.30	8.89
TTS + Real + SA Merge	8.84	8.85	8.84	8.74

3) *Comparison of Model Merging Techniques for Self Supervised SFMs*: For completeness, we also examine the effectiveness of merging methods on different Self Supervised SFMs in Table IV. In line with the findings of [10], we note a general trend of self-supervised SFMs exhibiting higher WER on child ASR tasks compared to their supervised counterparts. We also observe that task vector-based methods (including SA Merge) underperform direct parameter merging techniques like Lerp, although they still outperform fine-tuned models without any merging. We hypothesize that this discrepancy may be attributed to the significant task shift from the SSL objective to the CTC objective during the fine-tuning of Self Supervised SFMs. However, a more in-depth exploration of this phenomenon is left for future work.

TABLE IV

WER RESULTS ON MYST TEST SET FROM MERGING SELF SUPERVISED SFMS TRAINED ON MYST SUBSETS AND LS TRAIN-100 USING LERP AND SA MERGE. BOLD FACE NUMBERS INDICATE BEST RESULTS.

Model	Merging Method	MyST test WER			
		MyST 1hr	MyST 5hr	MyST 10hr	MyST full
Wav2Vec2.0	ft	31.95	21.05	18.43	13.18
	Lerp	30.40	19.87	17.77	12.83
	SA Merge	31.33	20.57	19.90	13.14
HuBERT	ft	34.78	21.81	19.29	13.30
	Lerp	32.09	20.60	18.07	13.01
	SA Merge	32.47	20.65	18.09	12.78
WavLM	ft	27.37	18.03	15.86	12.38
	Lerp	25.84	16.56	14.93	12.08
	SA Merge	26.52	17.41	15.28	12.23

B. Can we isolate Task Vectors from different data augmentation techniques?

1) *Transferability of Data Augmentation Task Vectors*: In addition to our investigations into the direct merging of models, we also intend to compute the task vectors for models that have been fine-tuned using various data augmentation techniques, as discussed in Section II-C. Subsequently, we investigate whether these computed task vectors can be effectively transferred to other datasets to enhance their performance. Different from [30], we do not compute an overall task vector based on performance across all tasks; instead, we focus on deriving task vectors specific to the particular dataset.

We begin by fine-tuning Whisper-small models on the entire MyST train corpus, incorporating common augmentations used in child ASR (PP, SP, SA, VTLP), as well as synthetic data. Subsequently, we compute the difference in task vectors between these models and transfer these differences to two Whisper-small models: one in a zero-shot setting and another fine-tuned on the CMU Kids corpus. Our results, presented in Table V, reveal that task vector transfer leads to a relative WER reduction of 18% in the zero-shot setting and 11% in the fine-tuned model, eliminating the need for retraining on augmented data for the new dataset.

2) *Alignment of Data Augmentation Task Vectors*: Building on the notion that task vectors encapsulate both task-specific information (which boost performance) and inherent robustness, we analyze the pairwise cosine similarities between the computed task vectors, following a similar approach to [30].

Our analysis in Figure 1 reveals that task vectors derived from conventional signal processing-based data augmentation techniques exhibit high similarity, suggesting that the performance gains from combining these techniques might be limited, as observed by [10]. In contrast, the low alignment of vectors from synthetic data indicates that this method could potentially be used in conjunction with other techniques to further enhance performance.

TABLE V
WER ON CMU KIDS TEST SET FROM TRANSFERRING TASK VECTORS (TV) OF AUGMENTATIONS (PP, VTLP, SP, SPECAUG, TTS) FROM WHISPER-SMALL FINE-TUNED ON MYST TO BOTH ZERO-SHOT (NO FINE-TUNING) AND CMU KIDS FINE-TUNED MODELS. BOLD FACE NUMBERS INDICATE BEST RESULTS.

Merging Method	CMU Kids test WER	
	zero-shot	fine-tuned
None	11.36	2.01
VTLP tv	9.52	1.91
PP tv	9.66	1.79
SpecAug tv	9.54	1.83
SP tv	9.48	1.86
TTS tv	9.29	1.85

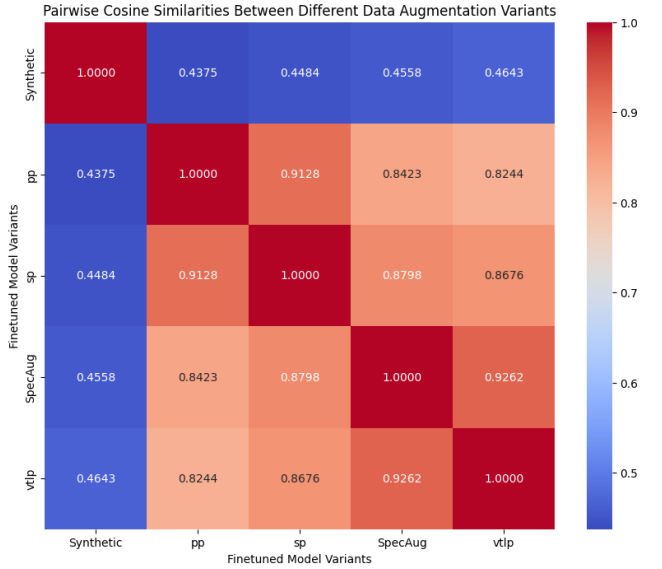


Fig. 1. Pairwise cosine similarities between task vectors derived from Whisper-small models fine-tuned on various MyST data augmentations.

V. CONCLUSION

In this paper, we explored the potential of model merging to enhance low-resource child automatic speech recognition (ASR) without the need to retrain models on new datasets. We introduced Selective Attention (SA) Merge, a novel technique that selectively merges task vectors from attention matrices to improve the performance of speech foundation models (SFMs) on child speech data. Our experiments demonstrated that model merging, particularly SA Merge, leads to significant improvements in low-resource scenarios, achieving relative word error rate (WER) reductions of up to 14%. By combining data augmentation techniques with SA Merge, we achieve a new state-of-the-art WER of 8.69 on the MyST database for the Whisper-small model. Further analysis of task vectors revealed the transferability of learned augmentations across datasets and the potential for combining multiple data augmentation techniques to further enhance child ASR systems. These findings underscore the efficacy of model merging, and SA Merge in particular, as a promising approach for addressing the challenges of child ASR in resource-constrained environments. Future research will explore the effectiveness of model merging in other low-resource domains, expanding its potential benefits beyond child ASR.

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